2011 F=ma Exam: Problem 25

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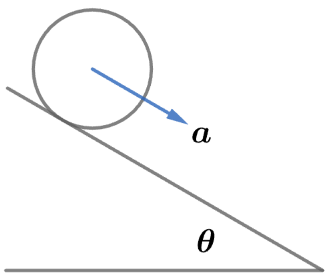
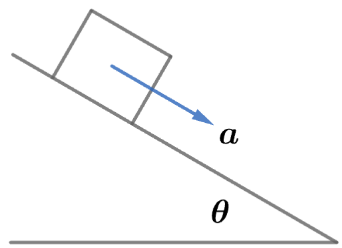
Recall the acceleration of an object rolling without slipping down an inclined plane is given by

$$a = \frac{g \sin\theta}{1+\beta}$$

where the moment of inertia of the object is $I = \beta m r^2$. Recall the acceleration of a block sliding down an inclined plane is given by

$$a = g(\sin\theta - \mu\cos\theta)$$





Since the hollow cylinder and block reach the bottom at the same time, they both have the same acceleration:

$$\frac{g \sin\theta}{1+\beta} = g(\sin\theta - \mu\cos\theta)$$

$$\mu\cos\theta = \sin\theta\left(1 - \frac{1}{1+\beta}\right)$$

For a hollow cyclinder, $\beta = 1$:

$$\mu\cos\theta = \frac{\sin\theta}{2}$$

$$\mu = \frac{\tan\theta}{2}$$

so the answer is $\boxed{\text{C}}$.